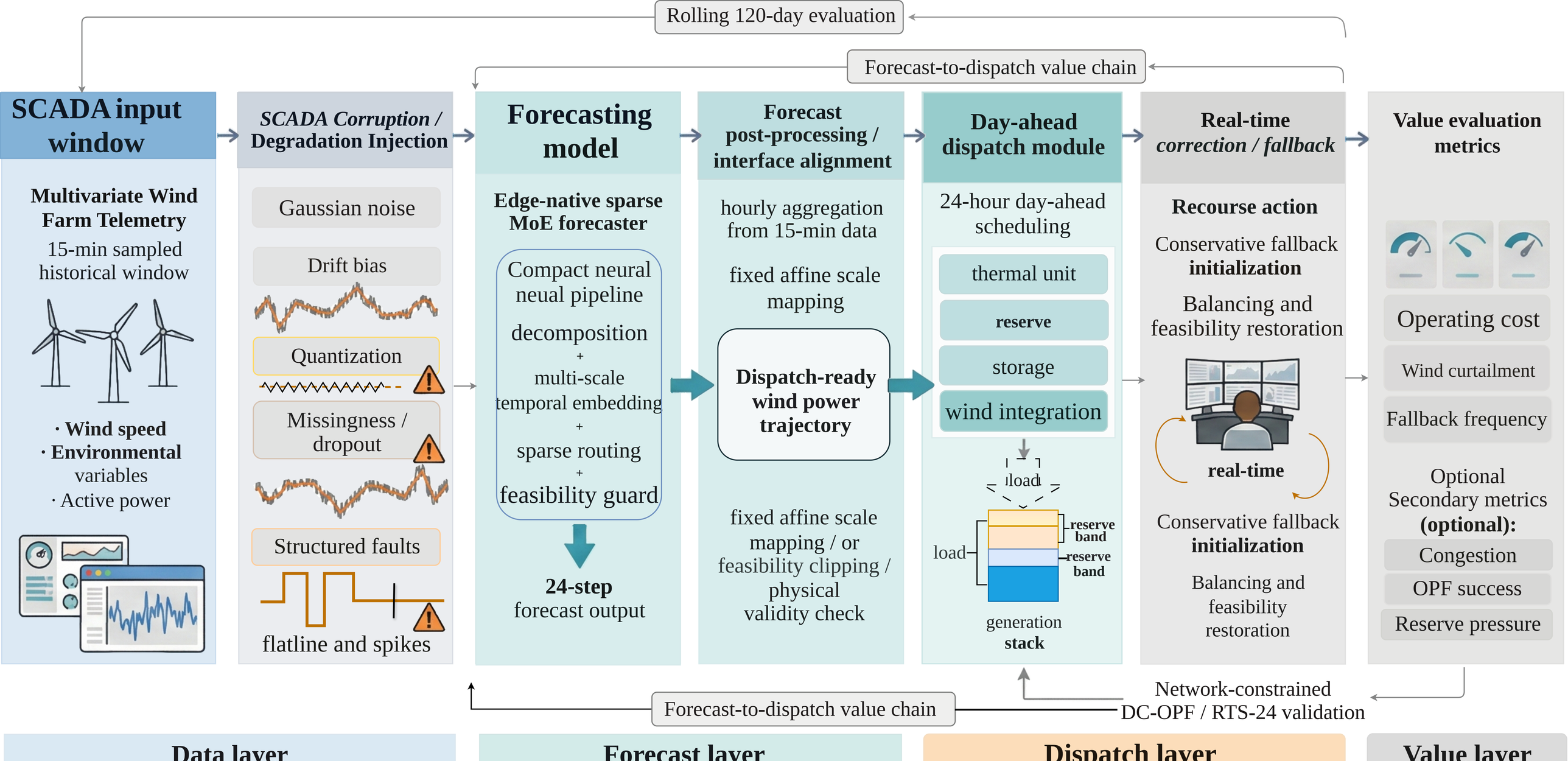
Rolling 120-day evaluation
Forecast-to-dispatch value chain
SCADA input window
Multivariate Wind Farm Telemetry
15-min sampled historical window
· Wind speed
· Environmental variables
· Active power
SCADA Corruption / Degradation Injection
Gaussian noise
Drift bias
Quantization
Missingness / dropout
Structured faults
flatline and spikes
Forecasting model
Edge-native sparse MoE forecaster
Compact neural neual pipeline
decomposition
+
multi-scale temporal embedding
+
sparse routing
+
feasibility guard
24-step forecast output
Forecast post-processing / interface alignment
hourly aggregation from 15-min data
fixed affine scale mapping
Dispatch-ready wind power trajectory
fixed affine scale mapping / or feasibility clipping / physical validity check
Day-ahead dispatch module
24-hour day-ahead scheduling
thermal unit
reserve
storage
wind integration
load
load
reserve band
reserve band
generation stack
Real-time correction / fallback
Recourse action
Conservative fallback initialization
Balancing and feasibility restoration
real-time
Conservative fallback initialization
Balancing and feasibility restoration
Value evaluation metrics
Operating cost
Wind curtailment
Fallback frequency
Optional Secondary metrics (optional):
Congestion
OPF success
Reserve pressure
Network-constrained DC-OPF / RTS-24 validation
Forecast-to-dispatch value chain
Data layer
Forecast layer
Dispatch layer
Value layer